Healthcare Patient Dataset

Visual report: age, outcomes by gender, and clinical correlations

600	5	50.99	29.0%
patients	diagnoses	mean age (18-88)	30-day readmission

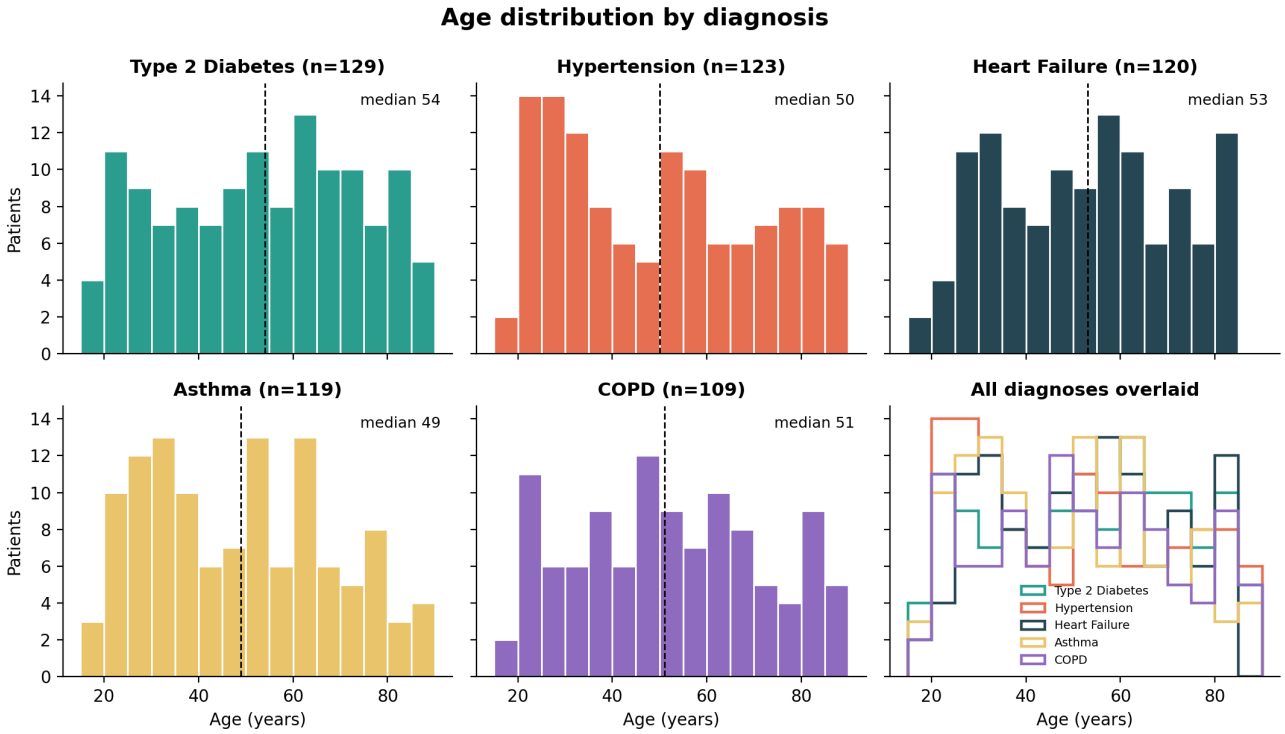
Key findings

- **Age is similar across diagnoses.** Median age ranges from 49 (Asthma) to 54 (Type 2 Diabetes); the difference is not statistically significant (Kruskal-Wallis p = 0.32).
- **Readmission differs by primary medication / diagnosis.** Tiotropium (COPD) 39% and Furosemide (Heart Failure) 36% are highest; Metformin (Type 2 Diabetes) is lowest at 19% (chi-square p = 0.003).
- **Gender differences are small and not significant.** Overall readmission: Male 33%, Female 29%, Non-binary 24% (p = 0.14). Subgroup sizes are only 26-48 patients, so gaps like Amlodipine-Male 38% vs Female 21% should be treated as exploratory.
- **Prior admissions are the strongest signal.** Readmission rises from 14% (0-1 prior admissions) to 48% (5 prior admissions); r = 0.29. Length of stay is weakly related (r = 0.13).
- **Age, comorbidity count and follow-up booking show no meaningful link to readmission** ($|r| < 0.04$, all $p > 0.4$). Follow-up rates are near-identical across genders (70-73%).

Dataset and method

The file contained 600 rows and 11 columns with no missing values and no duplicate PatientIDs, so no cleaning was required. Patient IDs are prefixed SYN-PAT, indicating **synthetic data**; findings illustrate the analysis and should not be read as clinical evidence. The dataset has no dedicated treatment-outcome field, so **outcome** is defined here as 30-day readmission (primary), with discharge disposition and follow-up booking as secondary outcomes. Each diagnosis maps to exactly one primary medication, so medication effects cannot be separated from diagnosis effects.

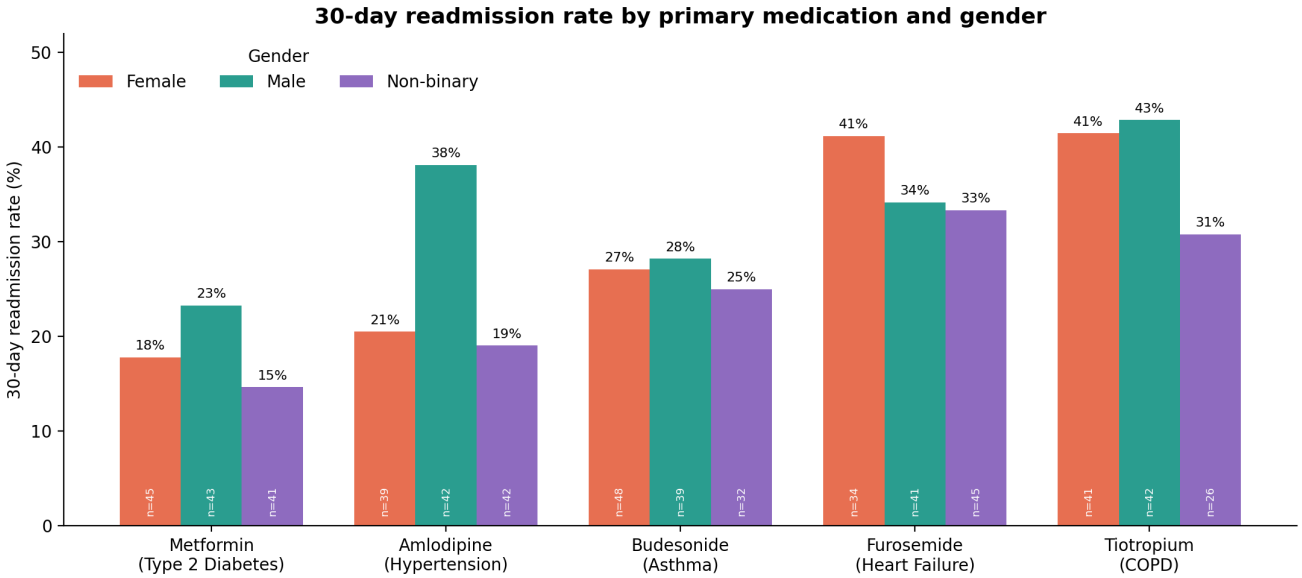
1. Age distribution by diagnosis



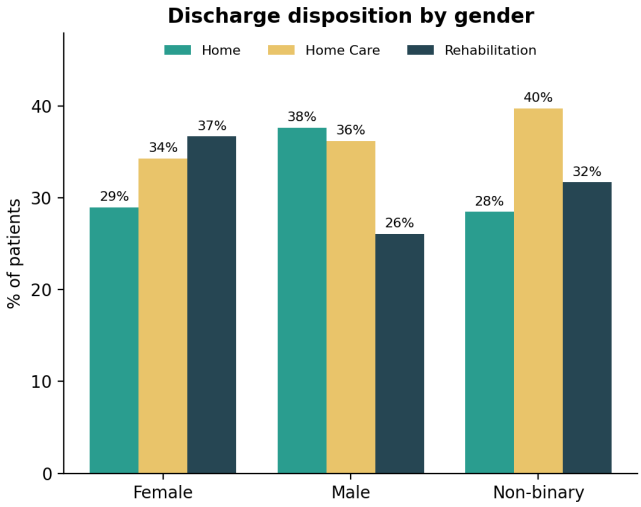
Diagnosis	n	Mean	Median	SD	Range
Type 2 Diabetes	129	53.0	54	20.2	18-88
Hypertension	123	49.7	50	21.3	18-88
Heart Failure	120	52.2	53	18.8	18-84
Asthma	119	48.3	49	19.2	18-88
COPD	109	51.8	51	20.0	18-87

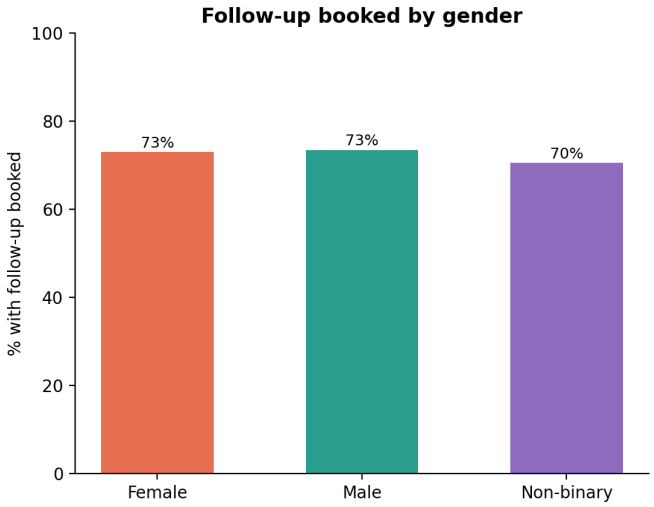
Every diagnosis spans adults from 18 to the mid-to-late 80s with a broad, roughly flat distribution rather than a clear peak. Dashed lines mark each group's median.

2. Treatment outcomes by gender



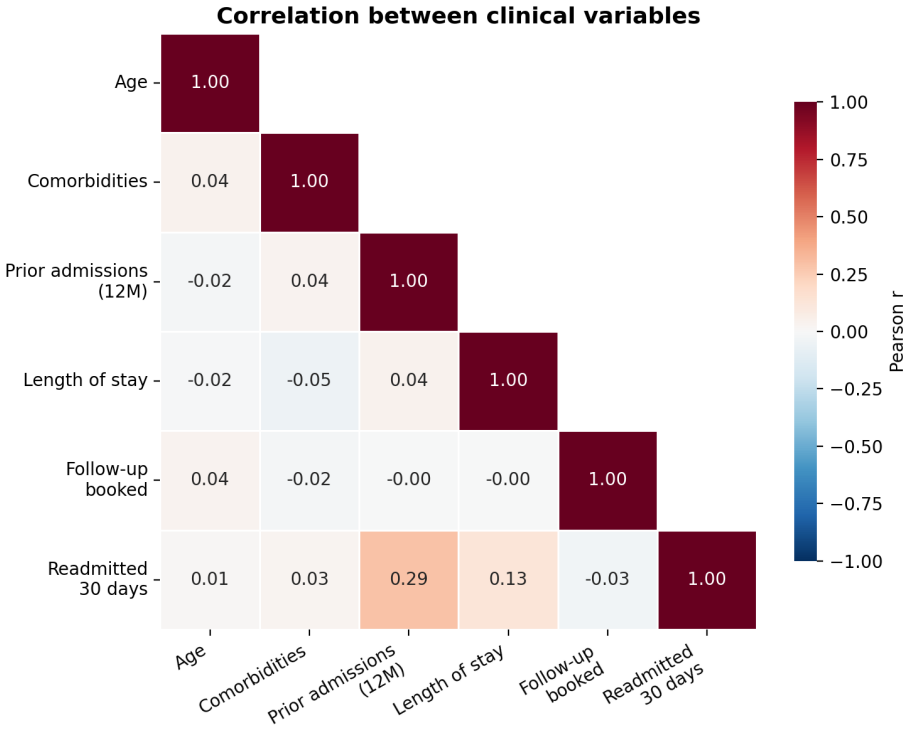
Bars show the share of patients readmitted within 30 days. White labels at the base give the group size. Medications are ordered by overall readmission rate. Small groups (e.g. Tiotropium, Non-binary, n = 26) make individual bars imprecise.





Males were more often discharged straight home (38%) while Females were most often sent to rehabilitation (37%); this pattern is borderline (p = 0.09). Follow-up booking does not differ by gender (p = 0.78).

3. Correlations between clinical variables



Pearson correlations (lower triangle shown). Follow-up booked and Readmitted are coded 0/1. The predictors are almost uncorrelated with one another, so there is little collinearity. Readmission correlates with prior admissions ($r = 0.29$, $p < 0.001$) and length of stay ($r = 0.13$, $p = 0.001$); other relationships are near zero.

Limitations

Correlation does not imply causation. Group sizes by gender and medication are small, no multiple-comparison correction was applied, and the data appear synthetic. Medication and diagnosis are fully confounded, so differences between medications reflect the underlying conditions as much as any drug effect.